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World First ESP-Ready ERD Smart Completion – Innovative Solutions to Reduce Future Well Interventions

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Abstract

Extreme Reach Drilling (ERD) has gained significant traction in oil and gas industry due to the advantage of accessing hydrocarbon reserves from a single wellbore, thereby enhancing reservoir contact and resources efficiency. However, extreme reach wells targeting multiples reservoir sections with flow control and monitoring system are complex to be completed. Reservoir management with minimal well interventions is always a challenge. Hence, the Completion design must be engineered from the start to help optimize the resources, simplifying the processes and extending the asset life. This paper will describe a successfully implemented completion design that enables future installation and replacement of artificial-lift methods, with minimal intervention and without retrieving the complex flow-control and monitoring system.

The significant challenge in completing extreme reach wells is ensuring that the completion can reach the target depth. Therefore, identifying appropriate friction factors to be used as basis of design is essential. Implementing recommended measures such as mechanical friction reducers, lubricants and strengthening the completion string is crucial. A tapered string design with different tubing weight was a key strategy to reduce the buckling risks in order to reach the target depth, but this required some wellhead modifications compared to standard operations. In addition, the selection of tapered string design was needed to meet the objective of allowing future installation and replacement of artificial-lift methods with minimum intervention. The completion design was continuously refined during the feasibility study stage to meet the well objectives, and the deployment procedures were meticulously prepared to ensure successful execution.

The successful deployment of the smart completion with ESP-ready capabilities in an extreme reach well (ERD ratio of 5.2:1 MD/TVD) was achieved through the collaborative effort of a multi-disciplinary team. Through multiple iterations of torque and drag simulations to identify the optimal pipe combination and modifications of existing products and system integrity tests, the team addressed the key best practices and benefited from many lessons learned from similar operations. This well became the world's first fully completed extreme reach well with the ability to install and replace an ESP with minimal intervention to de-complete and without the need to retrieve the well smart completion. This ability to install and replace artificial lift systems with minimal disruption not only supports long-term operational efficiency but also contributes to more sustainable production practices, ultimately reducing the environmental impact and

improving the cost-effectiveness. The valuable lessons learned will greatly contribute to refining the design and execution of future ERD projects in the oil and gas industry, ensuring more efficient, sustainable and successful outcomes.

Introduction

Extreme Reach Drilling (ERD) is defined as a wellbore that extends a significant horizontal distance from the surface location versus the total vertical depth. This provides access to subsurface targets that are far from the drilling rig. These wells are characterized by a high measured depth (MD) relative to their true vertical depth (TVD), often with a horizontal departure that exceeds twice the vertical depth (MD/TVD ratio: greater than 2:1). It is a field-proven and an economically attractive technical solution for unlocking opportunities in the oil and gas industry. ERD allows access of remote or multiple reservoirs from a single surface location, to reduce the environmental footprint and the infrastructure costs.

Drilling and completion of extreme reach wells are driven by surface location constraints. The only option to access the target reservoir is through a very challenging extreme reach trajectory well as shown in (Figure 1). This can lead to significant challenges for the completion, as working over the well in the future. Therefore, a comprehensive engineering study, supported by the collaborative efforts of a multidisciplinary team, was conducted to develop a very specific completion design tailored to meet the objective of controlling and monitoring the various zone, while adding the capability to later upgrade the artificial lift methods – such as Electric Submersible Pumps (ESP) – with minimal intervention to de-complete the well without retrieve the existing smart completion (ESP-Ready completion).

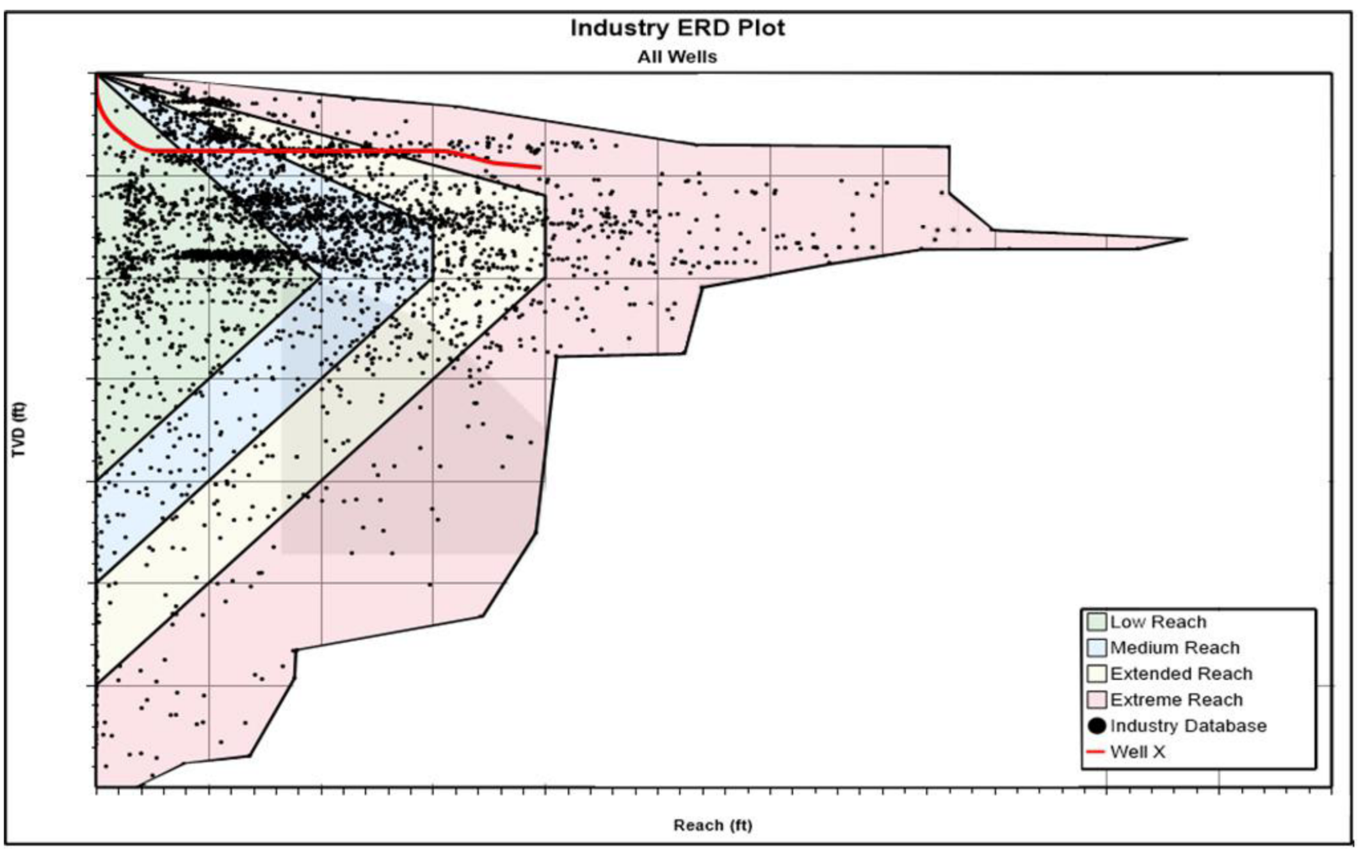

Figure 1—ERD Plot of Well X.

Challenges

The deployment of smart completion (consist of flow control and monitoring system) in extreme reach wells has numerous challenges, one of the biggest challenges is to successfully deploy the completion string to the target depth. With shallow true vertical depth (TVD) and long horizontal sections often exceeding 15,000 feet, drag and friction become significant obstacles during the installation of smart completion equipment and increase the risks of mechanical lock-up due to string buckling. Buckling occurs when the tubing experiences excessive axial loads, causing it to bend and potentially twist. Significant axial loads are required to push the completion string to reach target depth in ERD wells. As the string is forced through the high-friction horizontal section, it starts to buckle in the shape of helical or sinusoidal buckling. which leads to loss of completion weight and lock-up. (Figure 2) shows how buckling affects the effectiveness of hook-load in pushing the completion string.

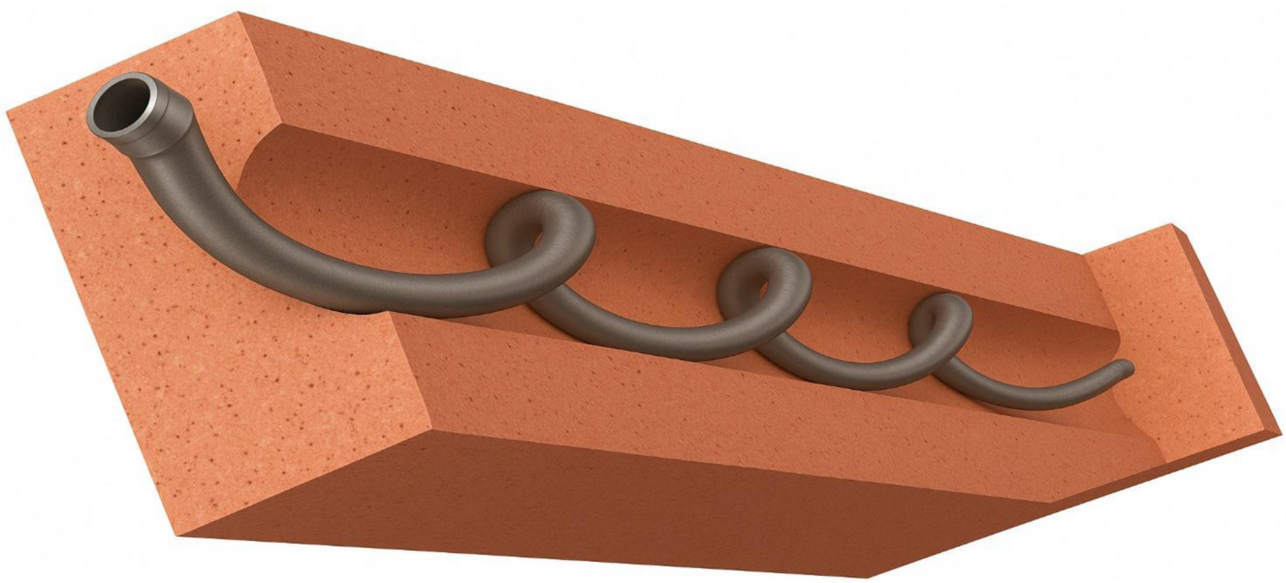

Figure 2—Buckling effect completion string causing string lock-up.

Rotation could be an easy solution to overcome the challenge of buckling and loss of completion weight. However, rotation is not possible while running smart completion because of the cables run on the side of the completion string and the packers sealing elements. Smart well completions are designed to allow operators to remotely control and monitor production from multiple compartments within a single bore. These completions typically include components such as interval control valves (ICVs), packers, sensors, and control lines (hydraulic, electric, or fiber optic), all of which are permanently installed during well completion. These control lines are carefully routed and secured along the production tubing and are not designed to withstand rotation. Rotating the tubing can cause the lines to twist, kink, or break, potentially leading to loss of communication with downhole tools or damage to sealing elements and packers, which can compromise zonal isolation and well integrity. As such, rotating a smart completion string can lead to damage to the system and undermine its functionality.

Drilling and completing Extreme Reach Drilling (ERD) wells are highly expensive due to the technical complexities and challenges involved. These wells are drilled to extreme depths and long horizontal lengths, requiring specialized equipment and advanced drilling and completions technologies which are more costly than conventional wells. The increased risk of wellbore instability, stuck pipe, and other drilling and completions deployment issues demands careful planning and more time, which drives up costs. Additionally, ERD wells require high-performance rigs, drilling fluids, and tools, as well as advanced completion systems to handle the long wellbores and ensure zonal isolation, monitoring and flow control.

Furthermore, all reservoirs naturally deplete over time as hydrocarbons are extracted and pressure declines. As the reservoir pressure drops, the ability for fluids to flow to the surface diminishes, and the natural drive mechanisms, like gas or water drive, become insufficient to maintain optimal production rates. At this stage, high-cost workover operations are required to de-complete and install artificial lift methods to sustain production. In this case installation of electrical submersible pumps (ESP)- become necessary to maximize recovery and enhance productivity. These factors make workovers in ERD wells both complex and expensive, underscoring the importance of making provision at the design and initial completion stage for installation of an artificial lift system to avoid costly well interventions in the future.

Given the complexity and the high cost of ERD wells, strategic planning during the design phase is essential to ensure long-term success and minimize operational risks. Mitigating ERD wells challenges require optimized completion string design, detailed pre-job torque and drag modeling and the use of specialized friction reducers to help the smart completion reach its intended depth reliably. Additionally, designing the well with future needs in mind such as the potential installation of an artificial lift system can significantly reduce the need for costly de-completion or interventions later, ultimately optimizing the long-term well performances and hydrocarbon recovery.

Completion Design

The well design as shown in (Figure 3) features a production casing and a non-cemented Liner consisting of open-hole metal expendable packers and pre-perforated liner across target reservoir. A Liner Hanger system consisting of a hanger, a liner top packer and an optimized tie back receptacle was used to safely deploy the open hole completion to depth while minimizing the risks when entering the liner with the smart completion.

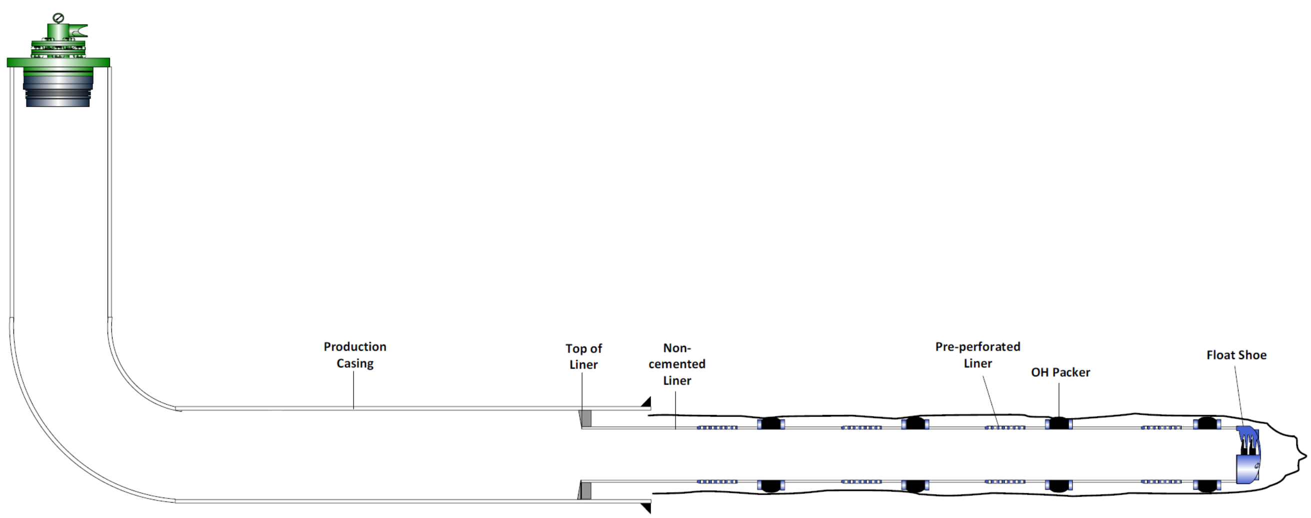

Figure 3—Lower Completion Design.

Tubing Selection

The selected tubing for the ESP-Ready ERD Smart Completion was based on a tapered string design. A tapered string completion uses a combination of different sizes of tubing, typically starting with larger diameter tubing at the top and gradually transitioning to a smaller diameter at the bottom. This design improves completion reachability by increasing buckling resistance and providing more weight in the vertical section to push the completion further in the long horizontal section. Due to the excessive loads that tubing will be subjected to in the ERD trajectory the tubing connection was selected based on the connection performance in tension and compression. Placement of each size of tubing was determined based on Torque and Drag simulations which validated an optimized completion size design. Heavy tubing in the vertical section was desirable in the vertical section for this ERD well but detrimental in the horizontal section, hence the tapers string design optimization was achieved with aid of simulation softwires and engineering judgment.

Smart Completion Tools and Accessories

To achieve the objectives of the ESP-Ready ERD Smart Completion, the system was designed for deployment in two separate runs. The first run includes the majority of the completion tools, such as blast joints with cable bypass, multi-port packers, interval control valves (ICV), and permanent downhole monitoring systems (PDHMS). These components are deployed as part of tubing within the non-cemented liner. Various tubing sizes are used within the production casing to provide additional weight in the vertical section with considering the ID restrictions for future interventions. The second run, referred to as the inner string, consists of a seal assembly, a sliding sleeve, and a subsurface safety valve. It is deployed on a smaller size of tubing inside the first run's tubing. This string can be easily retrieved, as it is relatively short, and it enables the future installation of an electrical submersible pump (ESP) inside the first run's tubing. The smart completion string (first run) will therefore remain in the wellbore for the life of the well.

ESP-Ready ERD Smart Completion First Run (Figure 4):

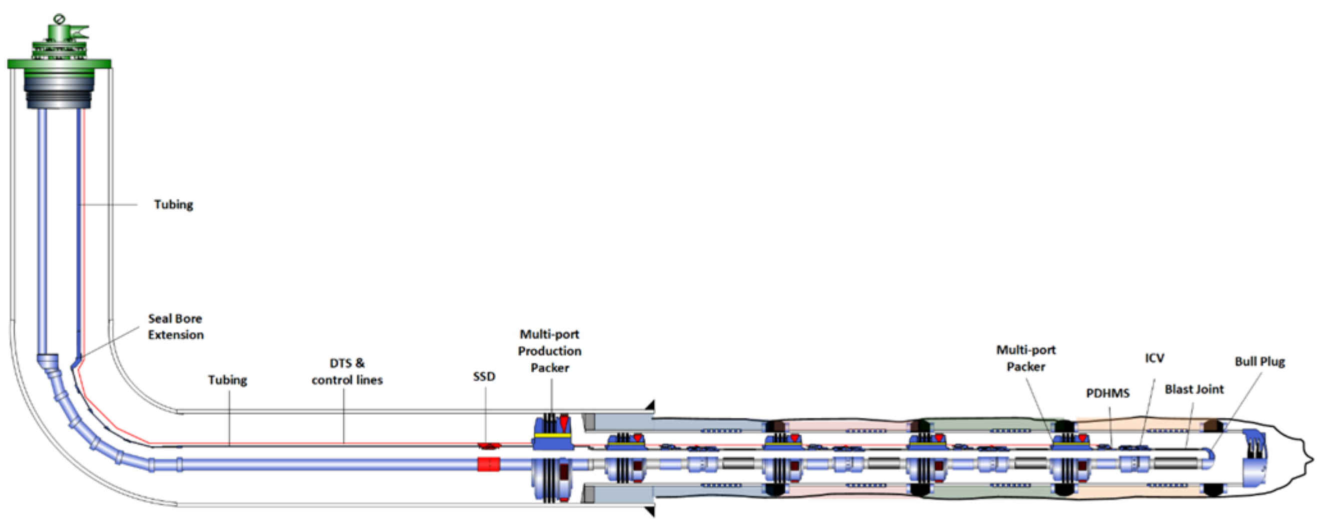

Figure 4—ESP-Ready ERD Smart Completion First Run.

Listed from bottom to top:

- **Bull-Nose Plug:** to plug end of completion to set packers against and to have smooth end to prevent the risk of hang-up when entering top of liner.
- **Blast Joints**: with control lines bypass, placed across the pre-perforated liner for control lines protection across flowing to prevent pipe erosion.
- **Interval Control Valves (ICV):** surface-controlled valve to control production in each compartment.
- **Permanent Downhole Monitoring System (PDHMS)**: pressure and temperature sensors for monitoring tubing and annulus in each compartment.
- **Multi-port Packers**: for anchoring and isolation (compartmentalization) with the feature of control lines bypass.
- **Multi-port Production Packer**: for anchoring and isolation above top of liner with the feature of control lines bypass.

- **Sliding Sleeve (SSD)**: placed above the production packer for well displacement.
- **Distributed Temperature Sensor (DTS)**: placed above the production packer to surface, utilized for casing leak detection.
- **Seal Bore Extension**: to provide sealing area for the second run seal assembly.
- **Tubing Hanger**: first run tubing hanger with 8 ports for control lines feedthrough.

ESP-Ready ERD Smart Completion Inner String -Second Run- (Figure 5):

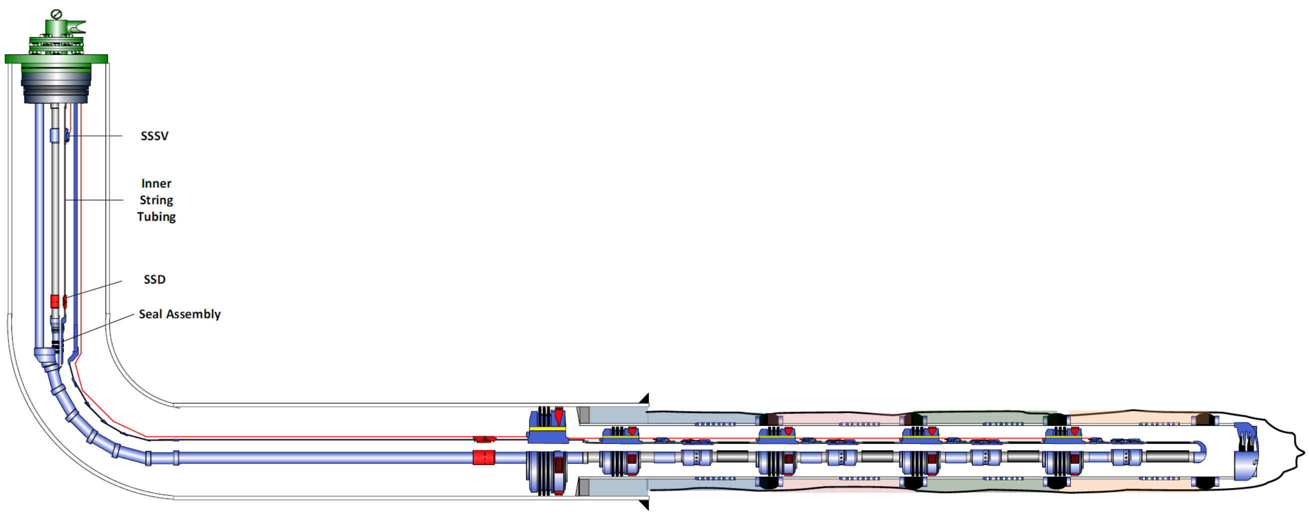

Figure 5—ESP-Ready ERD Smart Completion Inner String (Second Run).

Listed from bottom to top:

- **Seal Assembly:** to plug end of completion to set packers against and to have smooth end to prevent the risk of hang-up when entering top of liner.
- **Sliding Sleeve (SSD)**: placed above production packer for well displacement.
- **Sub Surface Safety Valve**.
- **Tubing Hanger**.

Friction Reducers

Friction reducers, such as lubricants and roller control line protectors, significantly reduce the resistance between the completion string and casing, minimizing the drag and buckling during the installation process. This results in smoother movement of the completion string, especially in extreme reach wells where the risk of stuck pipe is elevated, while also preventing damage to sensitive downhole equipment.

Mechanical Friction Reducers

For the deployment of the ESP-Ready ERD Smart Completion, Roller Control Line Protectors (Figure. 6) are used across tubing, while standard control line protectors are used across the tubing in the vertical section.

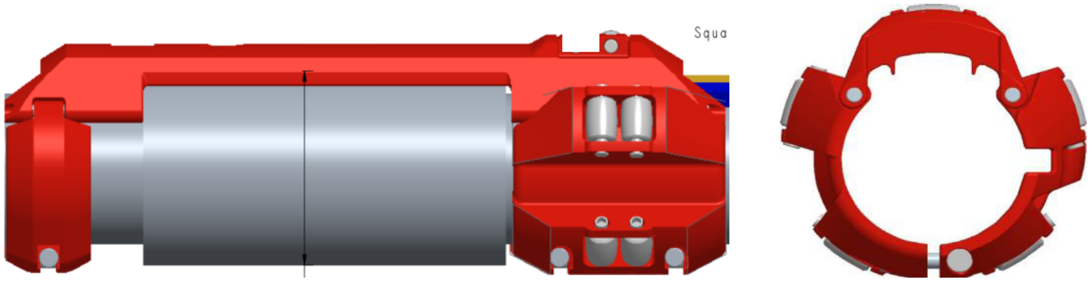

Figure 6—Roller Control lines Protectors.

Lubricant

Lubricants were stocked on the rig site as part of a contingency plan to address any unexpected increases in friction during the deployment of the completion string. These specialized friction-reducing additives were selected based on their compatibility with the wellbore environment and the completion fluids in use. In the event that the real-time monitoring indicated an elevated drag beyond predicted thresholds, the lubricants could be promptly mixed and pumped into the wellbore to reduce the drag and prevent potential lock-up. This proactive measure ensured operational flexibility and minimized the risk of non-productive time during running in hole.

Torque and Drag Modelling

The ability to reach target depth and set the completion equipment at the desired depth is highly dependent on the forces acting on the completion string and these forces are affected by a variety of factors. Effective Torque and Drag (T&D) modelling allows engineers to predict and mitigate potential operational issues related to reachability. Ensuring precision in Torque and Drag (T&D) modeling is vital, as incorrect inputs can produce unreliable outputs, potentially causing significant operational failures. Accurate modeling helps prevent these issues by providing a reliable basis for decision-making and ensuring successful execution.

The following best practices should be considered during Torque and Drag (T&D) modeling for an extreme reach well:

- **Accurate inputs:**
  Ensure precise measurement of well parameters, Include all relevant parameters such as mud properties, casing and tubing specifications, tool joints and actual well trajectory.
- **Conservative Assumptions:**
  Engineers should always take the conservative approach in friction factors assumption to ensure that the well completion will be able to handle unexpected conditions.
- **Sensitivity Analysis:**
  Perform sensitivity analysis to understand how variations in key parameters affect the model's outcomes, helping identify critical factors that impact T&D.
- **Hindcasting:** Historical data from completed wells should be used to validate friction factor assumptions, refining the model for future wells.
- **Real-time Calibration:**
  Real-time monitoring of torque and drag forces can provide valuable data for adjusting operations and facilitate decision making on spot, ensuring successful well completion.
- **Utilization of Advanced T&D Modelling Software:**
  The use of advanced modelling software equipped with robust computational algorithms is essential for achieving accurate, reliable and efficient analysis in planning and execution of these complex and extreme reach wells, potential risks can be identified early, design parameters can be fine-tuned, and decisions can be made to enhance reachability. In our experience, reliance on weak

or less sophisticated modelling software has led to significant discrepancies between predicted and actual outcomes. For instance, some basic models indicated that reaching the target depth (TD) was achievable within the operational constraints; however, during the actual deployment, the completion string could not be successfully run to TD.

Extensive Torque and Drag (T&D) modelling was conducted during the planning phase (Figure 7) of the ESP-Ready ERD Smart Completion. Several modifications were made throughout the process, incorporating hindcasting from similar wells to fine-tune the model's accuracy. Continuous monitoring of real-time operational data also played a crucial role in calibrating the assumptions. From the analysis of previous wells data, we concluded a set of friction factor assumptions that were used as the baseline for the Torque and Drag (T&D) modelling while planning new wells. These assumptions were selected using a conservative approach to ensure a safe operating margin, accounting for potential variations in downhole conditions and minimizing the risk of underestimating friction factors during actual deployment.

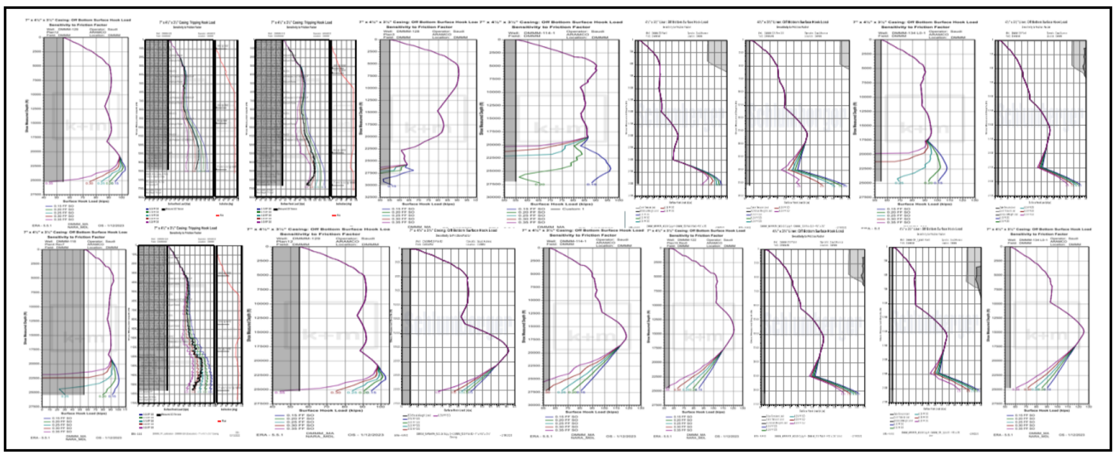

Figure 7—Examples of Torque and Drag Hindcasting and FF Calibrations.

ESP-Ready ERD Smart Completion (1st Run) WITHOUT lubricated brine:

- Production Casing Friction Factor: 0.20
- Liner Friction Factor: 0.35
- Roller Control lines Protectors FF Reduction: 45%

ESP-Ready ERD Smart Completion (1st Run) WITH lubricated brine:

- Production Casing Friction Factor: 0.15
- Liner Friction Factor: 0.15
- Roller Control lines Protectors FF Reduction: 45%

Below table is ESP-Ready ERD smart completion actual FF from XX wells:

Well	Production Casing FF	Liner FF
X1	0.18	0.32
X2	0.19	0.30
X3	0.20	0.34

Wellhead Modifications

To achieve the objective of enabling future ESP installations with minimal de-completion and intervention, the wellhead equipment was strategically modified to incorporate a dual tubing hanger system. This design allows for the installation of two tubing hangers (figure 8), one for the ESP-Ready smart completion first run (first run Tubing Hanger) and another to accommodate the inner string (second run Tubing Hanger). This configuration significantly simplifies the retrieval of the inner string and facilitates the installation of the ESP inside the production tubing, reducing operational complexity and overall intervention costs.

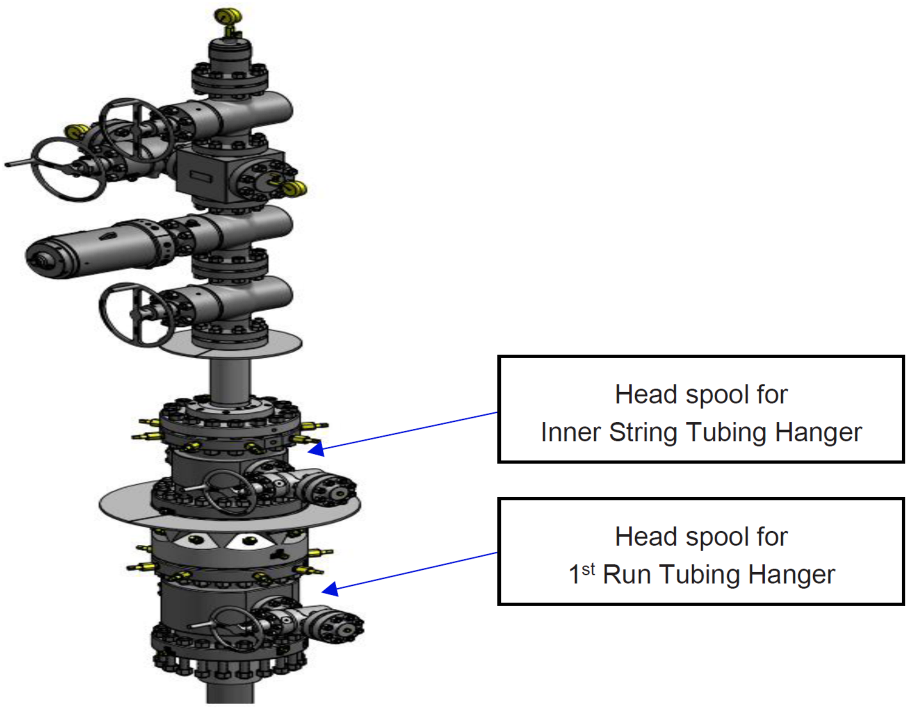

Figure 8—Well Head equipment.

Preparation Best Practices

Listed below are some of the best practices that are important to consider when preparing for an ESP-Ready ERD Smart Completion, particularly due to the unique challenges posed by the ERD well trajectory. These practices help ensure deployment success in such complex environments.

Wellhead System Integration Test (SIT)

Given that smart completions require multiple control lines to actuate downhole tools and considering that modifications were made to the wellhead equipment to accommodate the ESP-Ready ERD Smart Completion, it was essential to perform a system integration test at an early stage. This test ensured the successful feedthrough of smart completion control lines through the modified wellhead components (figure 9), verified the proper assembly of the wellhead equipment, and confirmed the integrity of the control lines under simulated operational conditions. Performing the wellhead SIT early not only mitigated the risk of last-minute complications but also allowed the team to capture the best practices and lessons learned, which would be critical for ensuring a smooth and efficient execution while deploying the completion.

Figure 9—Wellhead System Integration Test (SIT).

Passing Top of Liner System Integration Test (SIT)

Passing through the top of the liner with a smart completion string is often a critical concern, particularly in extreme reach wells where the top of the liner is situated within the horizontal section. In such cases, gravity causes the completion assemblies to rest along the low side of the hole, which increases the risk of the string hanging up at the liner top due to the restriction in internal diameter (ID). This becomes especially challenging when deploying complex smart completion assemblies that include multiple control lines.

To mitigate this risk, it is essential to conduct a "Passing Top of Liner" test early in the planning and preparation phase. This test helps to verify that each component of the smart completion can successfully and smoothly pass through the liner top. The test is typically performed in a workshop environment using a setup that simulates actual downhole conditions. In this case, a joint of casing with a machined window was secured in a vice, and the liner hanger assembly was installed inside. Smart completion assemblies including control line protectors, rollers, and other accessories were then manually pushed through the liner top in various orientations to observe their interaction with the restriction (Figure 10).

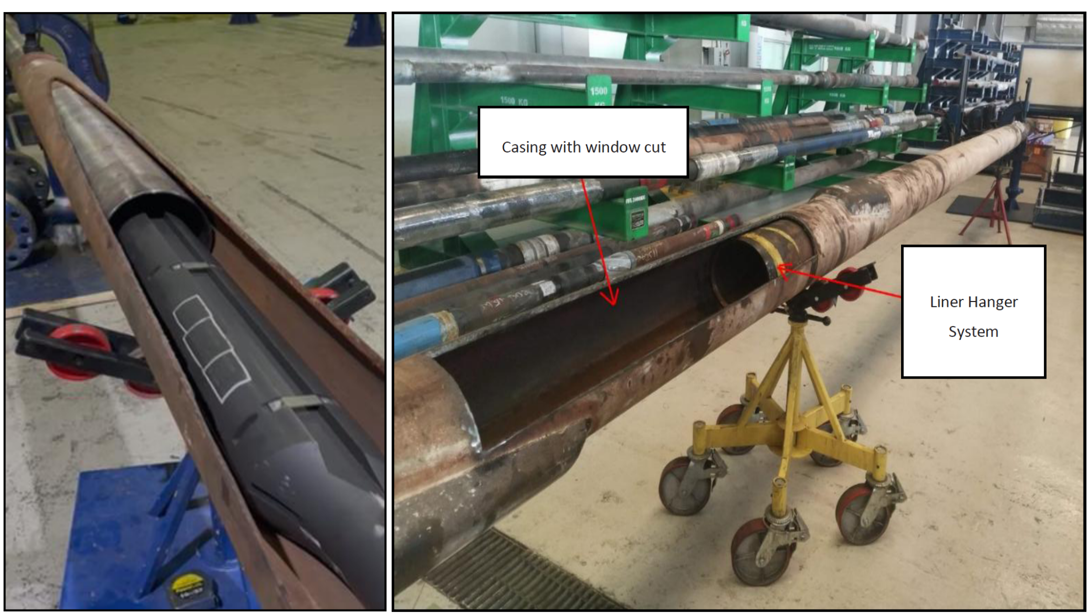

Figure 10—Passing Top of Liner System Integration Test (SIT).

After performing the test with the ESP-Ready ERD Smart Completion assemblies, we identified a couple of potential hanging points (Figure 11), particularly at the multi-port packers and within the blast joint assembly. these increased the risks of a stuck completion. To address this, design modifications were made: a bottom guide sub was added to the base of the packer to protect the control lines and to guide the assembly through the liner top smoothly. Additionally, the blast joint assembly was redesigned by removing the 90-degree shoulders, which were identified as potential hanging points. These design improvements significantly reduced the risk of mechanical interference and enhanced the overall reachability of the ESP-Ready ERD Smart Completion.

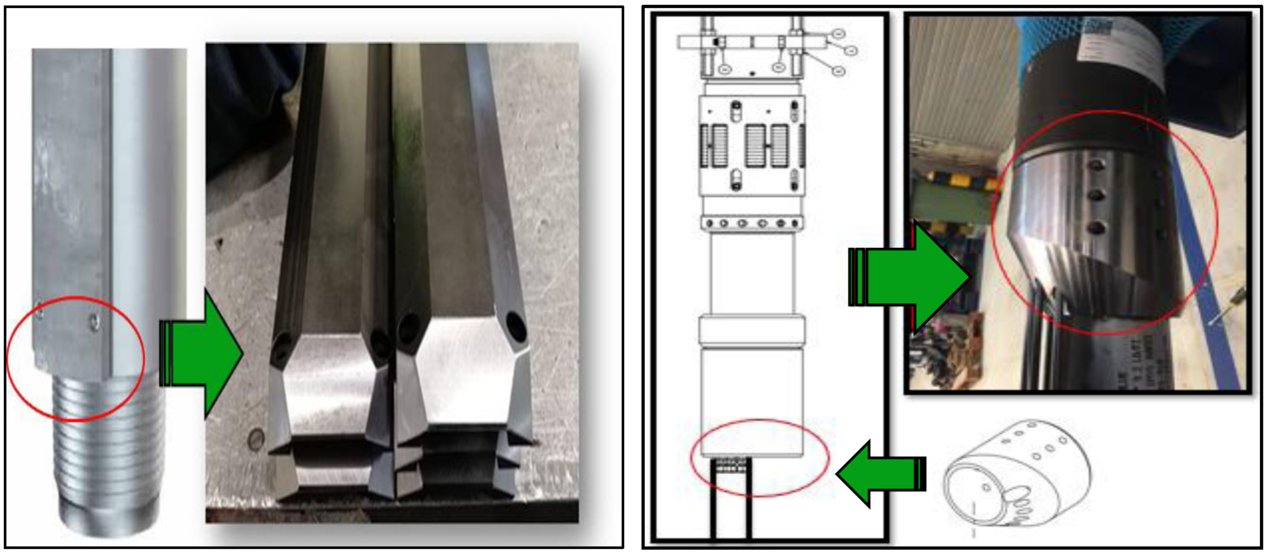

Figure 11—Modifications made after Passing Top of Liner System Integration Test observations.

Electrical System Integration Test (SIT)

As part of the smart completion system, the Permanent Downhole Monitoring System (PDHMS) includes electrical sensors designed to continuously record pressure and temperature in each production zone for effective real-time monitoring. While power supply and data transmission between surface and downhole sensors are generally straightforward in conventional wells, they pose significant challenges in Extreme Reach Drilling (ERD) environments due to the increased well length and the potential signal degradation. To address these challenges and to ensure the long-term system reliability, conducting an Electrical System Integration Test (SIT) is critical in every well. This test verifies that power and data signals can be transmitted effectively from the surface acquisition unit to all downhole sensors, remaining within the system design limits. The electrical SIT also helps to identify any potential issues with signal loss, grounding, or electrical resistance that could compromise the sensor performance during the deployment and the life of the well.

To simulate the worst-case scenario where the electrical cable remains on the reel with minimal grounding, as it would after installing the first sensor while running the completion the test was carried out in a workshop environment. All PDHMS sensors were connected to the full length of cable wound on the reel (Figure 12), and the system was then connected to the surface acquisition unit. Once powered, electrical reference parameters were measured and recorded. This approach provided a comprehensive validation of the system health, significantly reducing the risks of data loss or sensor failure during running in hole and over the life of the well.

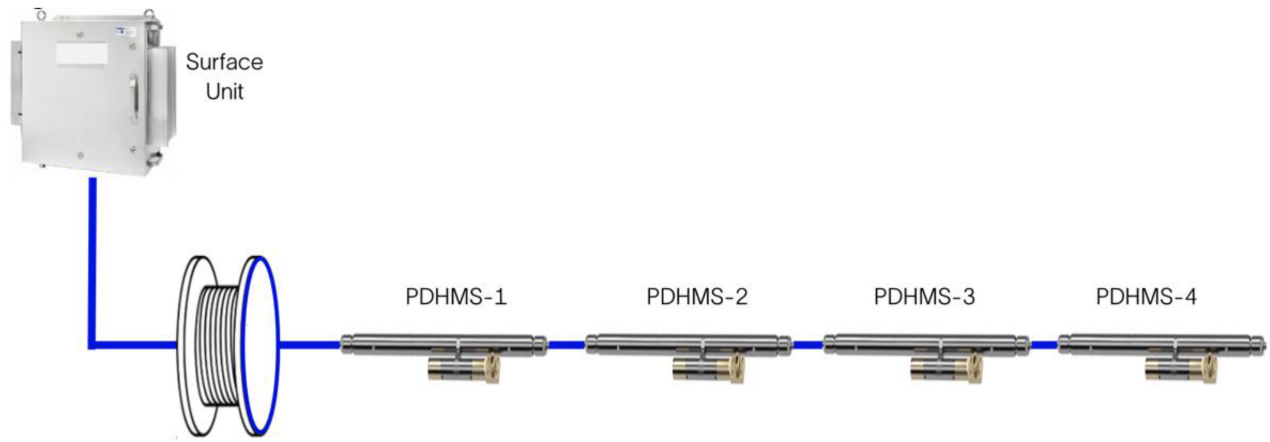

Figure 12—Electrical System Integration Test (SIT).

Eccentric Assemblies Alignment

When the completion assemblies are made up together, they become as a rigid assembly, making it especially important to carefully manage the orientation of eccentric components such as multiport packers and PDHMS mandrels. If these assemblies are made up with their high sides facing in opposite directions, it can significantly increase the overall outer diameter (Figure 13). This misalignment can lead to the assembly getting stuck while running into the restricted ID of the liner, particularly in extreme reach wells where excessive string buckling and side contact are common due to the long horizontal sections.

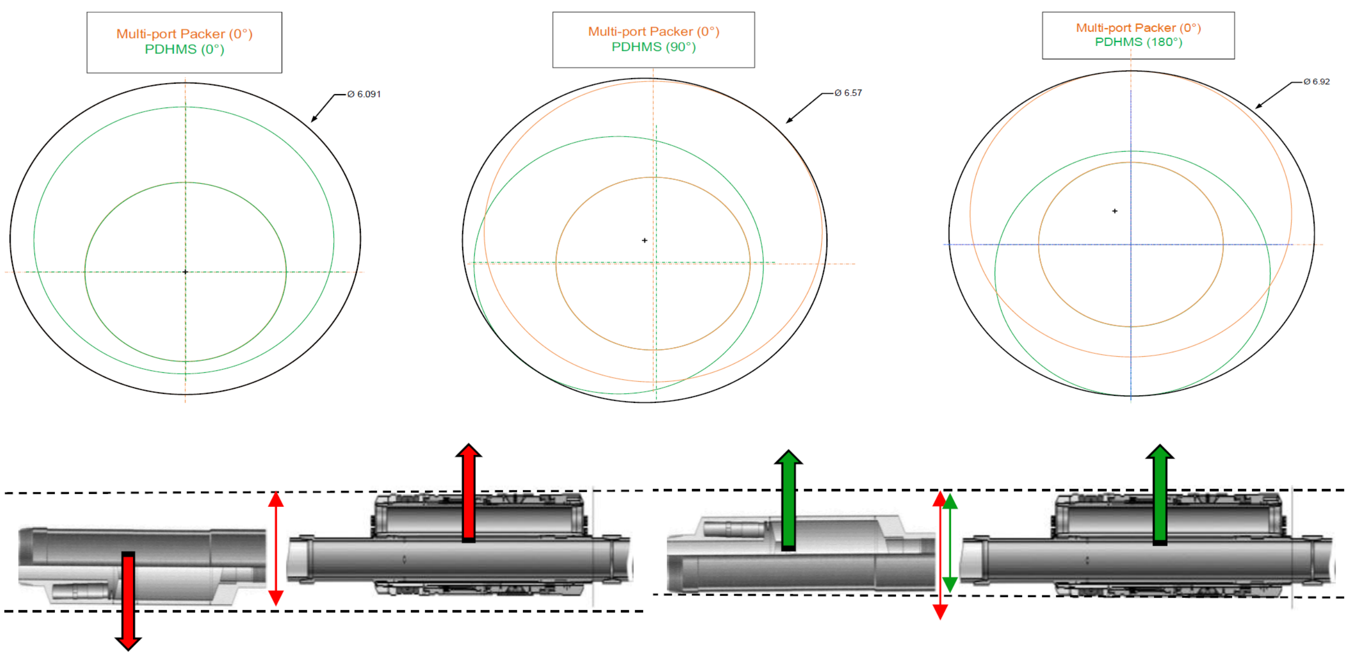

Figure 13—Explains how eccentric assemblies increased the overall running OD when it is not aligned.

To mitigate this risk, timed threads were used during assembly make up. Timed threads allow precise control of the rotational orientation of each component during make-up, ensuring that all the eccentric assemblies are aligned in the same direction. This practice helps to maintain a consistent and minimized overall combined OD along the completion string, reducing the likelihood of hang-ups.

Control-lines Management

Control lines whether hydraulic, electric, or fiber optic are essential components of a smart completion system, as they enable the actuation of downhole tools and the transmission of real-time data to surface. However, deploying these lines presents a significant challenge, particularly in Extreme Reach Drilling (ERD) wells. The complexity lies not only in routing and securing the lines along the completion string, but also in ensuring proper protection, and centralization throughout the well profile. In ERD environments, control lines that are run along the tubing are particularly vulnerable. If the completion string experiences severe buckling which is common in long horizontal sections there is a potential risk for the control lines to be pinched, crushed, or sheared. Such damage can lead to a total loss of communication or control to the smart downhole components, potentially compromising the entire smart completion system.

Therefore, meticulous planning and management of control lines is critical. Each line must be carefully mapped from its starting point on surface to its termination point downhole, with detailed attention given to its placement across various completion assemblies (figure 14). The selection and positioning of the control line protectors must be pre-defined, ensuring adequate protection at key transition points and areas of potential contact.

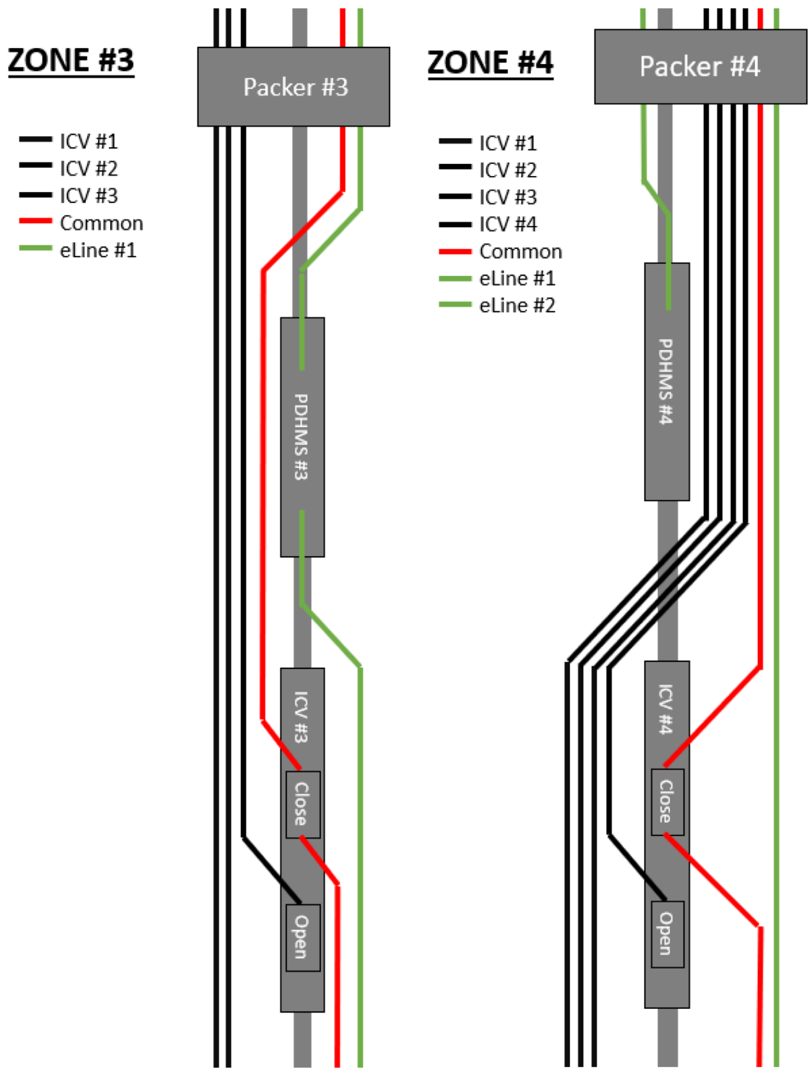

Figure 14—Example of Control line management across completions assemblies.

Additionally, proper layout and handling of control line spooling units at the rig site (Figure 15) should be established in advance, along with a clear plan for the use of sheave wheels to guide the lines safely into the wellbore during the deployment.

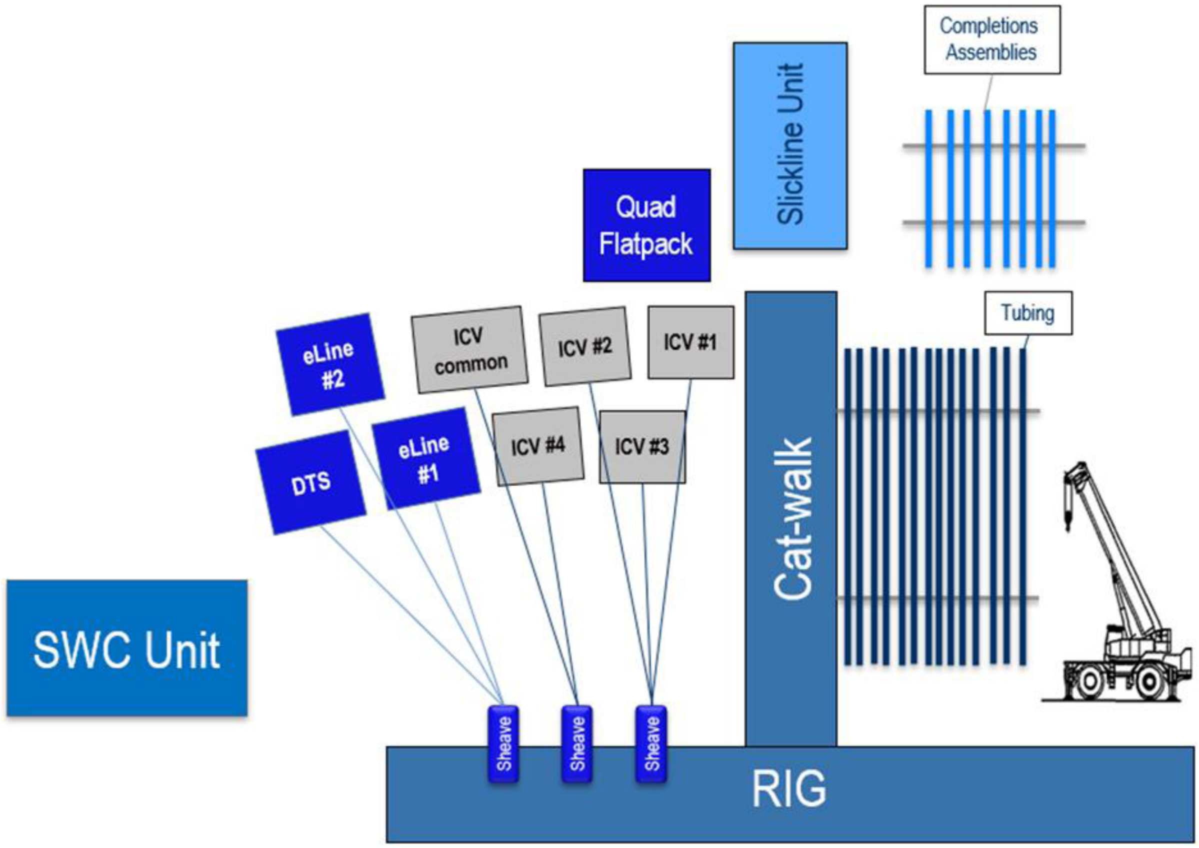

Figure 15—Example of Control lines spooling units management.

Deployment Risk Assessment

The table below presents general risk control measures tailored for ESP-Ready ERD Smart Completion. These measures are developed to proactively address and mitigate potential risks associated with the installation of smart completions in extreme reach wells. By incorporating both preventive and corrective controls, the measures aim to enhance system reliability and maintain the well integrity.

Table 1—Risk control measures Table.

Event	Details
Unable to reach TD (lost slack-off weight)	1. Attempt to work the string without slack-off more than 5 klbs 2. Circulate lubricated brine with 0.5% lubricant through each ICV and work the string 3. Increase lubricant percentage to 1% and circulate through each ICV and work the string 4. Pull out of hole and run wellbore clean out
Electrical line damage while running in hole	Pull back to damaged line and make an electrical splice • use Splice Protectors to protect the splice
Electrical line failure at sensor electrical connection	Remove the lower pup joint below PDHMS assembly to have extra length of electrical line and Re-do the sensor electrical connection
Setting Packers – tubing pressure not holding	Fully open interval control valves and re-close while circulating (1.5 BPM/150 psi) to clean sealing area
Multiport Packers – fail to set	Repeat setting procedure and hold pressure for longer time Increase pressure and hold pressure for 30 min

Event	Details
Damage lines while removing BOP	Make up lines wraps at tubing hanger neck to be use the Wraps made at TH as back up
Losses prior to running smart completion	In case losses are present prior to RIH smart completion, pump through plug is required to be installed below the packers: • Install Pump Thru Plug (PTP) above ICV #1. RIH completion until ICV #2. Fill up the string every so often. Monitor downhole pressures from the sensors to ensure no differential pressure. • Before picking up ICV #2, RIH slickline and retrieve PTP. • Install PTP above ICV #2. RIH completion until ICV #3. Fill up the string every so often. Monitor downhole pressures from the sensors to ensure no differential pressure. • Repeat steps for ICV #3, and ICV #4. • After picking up production packer, RIH slickline and retrieve PTP above ICV #4.

Successful Deployment

The first ESP-Ready ERD Smart Completion was successfully deployed in an extreme reach well, marking a significant milestone in advanced well completion technology. Following this initial success, multiple deployments were carried out across similar challenging environments.

Throughout these operations, the team encountered several technical challenges. These challenges provided valuable insights that enhanced our understanding of the deployment risks. Each experience contributed to the continuous refinement of our best practices, risk mitigation and operational procedure. As a result, we've been able to improve the reliability, efficiency, and safety of future installations. The lessons learned have not only helped reduce non-productive time and potential operational costs but also strengthened cross-functional collaboration between engineering and field operations.

This improvement process reinforces our commitment to operational excellence and continuous improvement delivering smart well completions in challenging trajectories that optimize production performance throughout the well lifecycle.

Cost Saving Factors

Utilizing ESP-Ready ERD smart completions delivers significant economic advantages throughout the life of a well, especially in late-life stages where artificial lift systems are required.

Key cost-saving benefits include:

- Reduced workover rig time costs.
- Avoid potential milling operations and associated costs.
- Prevention of casing damage during milling.
- Lower costs of additional mud due to unexpected losses during workover.
- Minimization of unnecessary production delays, enabling to put the wells on production faster.
- Reduced risk of formation damage during workover operations.
- Decreased likelihood of losing the well.
- Smart completion costs, including ICVs, control lines, protectors, packers, and related accessories.
- Lower additional tubing costs.

Summary and Way forward

More than 10 of ESP-Ready ERD Smart Completions have been successfully deployed in extreme reach wells. These installations have demonstrated the system's capability to perform reliably in some of the most technically demanding environments.

The way forward is extending the use of the ESP-Ready ERD Smart Completions in even more challenging wells, with an ERD ratio reaching up to 6.5 MD/TVD. Also, to simplify the smart completion architecture, a shift toward a fully electric solution will bring significant advantages like eliminating the complexity associated with multiple hydraulic control lines, reduce installation risks and enhancing the system reliability.

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